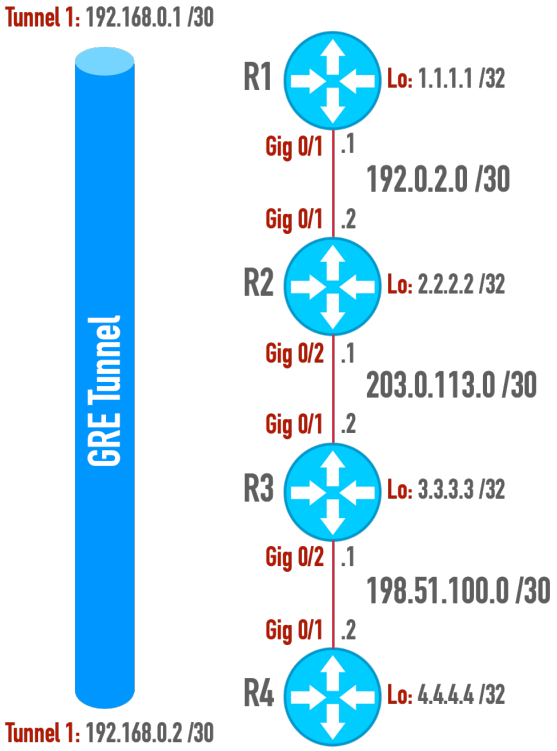
# GRE over IPsec

## Topology




## Initial Configuration Commands

### R1:

```
enable
conf t
host R1
no banner motd
no banner login
no banner exec
no banner incoming
line vty 0 15
password cisco
login
exec-timeout 0 0
line con 0
logging synchronous
exit
no ip-domain lookup
int lo0
```

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```
ip address 1.1.1.1 255.255.255.255
int gig 0/1
ip address 192.0.2.1 255.255.255.252
no shut
router ospf 1
network 0.0.0.0 0.0.0.0 area 0
end
copy run star
```

**R2:**

```
enable
conf t
host R2
no banner motd
no banner login
no banner exec
no banner incoming
line vty 0 15
password cisco
login
exec-timeout 0 0
line con 0
logging synchronous
exit
no ip-domain lookup
int lo0
ip address 2.2.2.2 255.255.255.255
int gig 0/1
ip address 192.0.2.2 255.255.255.252
no shut
int gig 0/2
ip address 203.0.113.1 255.255.255.252
no shut
router ospf 1
network 0.0.0.0 0.0.0.0 area 0
end
copy run star
```

**R3:**

```
enable
conf t
host R3
no banner motd
no banner login
```

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```
no banner exec
no banner incoming
line vty 0 15
password cisco
login
exec-timeout 0 0
line con 0
logging synchronous
exit
no ip-domain lookup
int lo0
ip address 3.3.3.3 255.255.255.255
int gig 0/1
ip address 203.0.113.2 255.255.255.252
no shut
int gig 0/2
ip address 198.51.100.1 255.255.255.252
no shut
router ospf 1
network 0.0.0.0 0.0.0.0 area 0
end
copy run star
```

**R4:**

```
enable
conf t
host R4
no banner motd
no banner login
no banner exec
no banner incoming
line vty 0 15
password cisco
login
exec-timeout 0 0
line con 0
logging synchronous
exit
no ip-domain lookup
int lo0
ip address 4.4.4.4 255.255.255.255
int gig 0/1
ip address 198.51.100.2 255.255.255.252
no shut
router ospf 1
```

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```
network 0.0.0.0 0.0.0.0 area 0
end
copy run star
```

## Lab Tasks

1. Let's set up just a basic GRE tunnel going from R1 to R4.
2. Let's take a look at the GRE tunnel that we just created.
3. Let's configure an IKE Phase 1 policy to be used by IPsec on R1.
4. Since we indicated we're going to be using a pre-shared key, we need to specify what the key is, and who it can be used by. For this lab, let's make the key "kevinskey", and let's make it available to any peer.
5. On R1, let's create a Transform set for our IKE Phase 2 connection. In this lab we are going to be using transport mode. Name our transform set "KWTRAIN".
6. Let's define our "interesting" traffic for the IPsec tunnel, and we want it to be our GRE traffic. Name our access control list "GRE-IN-IPSEC".
7. Let's create a Crypto Map and name it "VPN". This is going to be the thing that ties everything together, the interesting traffic, the transform set, and we point all this to our peer.
8. We now need to apply this crypto map to the gig 0/1 interface, which is the interface that we are going to use to send the GRE packets.
9. Let's now go to R4 and give a mirrored configuration of R1.
10. On R4, let's do some configuration verification by sending some traffic over the tunnel. Let's see if the IPsec tunnel is encrypting and decrypting GRE packets (our only interesting traffic).

## Solution

**Step 1:** Let's set up just a basic GRE tunnel going from R1 to R4.

```
R1>en
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#int tunnel 1
R1(config-if)#ip address 192.168.0.1 255.255.255.252
R1(config-if)#tunnel source gig 0/1
R1(config-if)#tunnel destination 198.51.100.2
R1(config-if)#end

R4>en
R4#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R4(config)#int tunnel 1
```

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```
R4(config-if)#ip address 192.168.0.2 255.255.255.252
R4(config-if)#tunnel source gig 0/1
R4(config-if)#tunnel destination 192.0.2.1
R4(config-if)#end
```

**Step 2:** Let's take a look at the GRE tunnel that we just created.

```
R1#show ip int brief
Interface                  IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0         unassigned      YES TFTP   administratively down down
GigabitEthernet0/1         192.0.2.1       YES TFTP   up                    up
GigabitEthernet0/2         unassigned      YES TFTP   administratively down down
GigabitEthernet0/3         unassigned      YES TFTP   administratively down down
Loopback0                  1.1.1.1         YES TFTP   up                    up
Tunnel1                    192.168.0.1     YES manual up                    up

R4#show ip int brief
Interface                  IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0         unassigned      YES TFTP   administratively down down
GigabitEthernet0/1         198.51.100.2    YES TFTP   up                    up
GigabitEthernet0/2         unassigned      YES TFTP   administratively down down
GigabitEthernet0/3         unassigned      YES TFTP   administratively down down
Loopback0                  4.4.4.4         YES TFTP   up                    up
Tunnel1                    192.168.0.2     YES manual up                    up

R4#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           0   FULL/  -        00:00:37    192.168.0.1     Tunnel1
3.3.3.3           1   FULL/BDR        00:00:31    198.51.100.1    GigabitEthernet0/1
```

*(##See how the GRE tunnel that we created is administratively up on both R1 and R4.)*
*(##On R4 it looks like we are physically adjacent, with R1, because of the GRE tunnel that we created.)*

**Step 3:** Let's configure an IKE Phase 1 policy to be used by IPsec on R1.

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#crypto ?
  call          Configure Crypto Call Admission Control
  ctcp          Configure cTCP encapsulation
  dynamic-map   Specify a dynamic crypto map template
  engine        Enter a crypto engine configurable menu
  gdoi          Configure GKM (Group Key Management, GDOI or G-IKEv2) Policy
  gkm           Configure GKM (Group Key Management, GDOI or G-IKEv2) Policy
  identity      Enter a crypto identity list
  ikev2         Configure IKEv2 Options
  ipsec         Configure IPSEC policy
  isakmp        Configure ISAKMP policy
  key           Long term key operations
  keyring       Key ring commands
  logging       logging messages
```

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```
  map            Enter a crypto map
  mib            Configure Crypto-related MIB Parameters
  pki            Public Key components
  provisioning   Secure Device Provisioning
  vpn            Configure crypto vpn commands
  wui            Crypto HTTP configuration interfaces
  xauth          X-Auth parameters

R1(config)#crypto isakmp policy 10
R1(config-isakmp)#encryption ?
  3des  Three key triple DES
  aes   AES - Advanced Encryption Standard.
  des   DES - Data Encryption Standard (56 bit keys).

R1(config-isakmp)#encryption aes
R1(config-isakmp)#authentication ?
  pre-share  Pre-Shared Key
  rsa-encr   Rivest-Shamir-Adleman Encryption
  rsa-sig    Rivest-Shamir-Adleman Signature

R1(config-isakmp)#authentication pre-share
R1(config-isakmp)#group ?
  1   Diffie-Hellman group 1 (768 bit)
  14  Diffie-Hellman group 14 (2048 bit)
  15  Diffie-Hellman group 15 (3072 bit)
  16  Diffie-Hellman group 16 (4096 bit)
  19  Diffie-Hellman group 19 (256 bit ecp)
  2   Diffie-Hellman group 2 (1024 bit)
  20  Diffie-Hellman group 20 (384 bit ecp)
  21  Diffie-Hellman group 21 (521 bit ecp)
  24  Diffie-Hellman group 24 (2048 bit, 256 bit subgroup)
  5   Diffie-Hellman group 5 (1536 bit)

R1(config-isakmp)#group 2
R1(config-isakmp)#exit
```

*(##We use isakmp to set up the IKE Phase 1 policy.)*
*(##We choose AES because it is the strongest of all three of these encryptions.)*
*(##The pre-shared key can be generated using Diffie-Hellman. Diffie-Hellman determines how secure the key is.)*

**Step 4:** Since we indicated we're going to be using a pre-shared key, we need to specify what the key is, and who it can be used by. For this lab, let's make the key "kevinskey", and let's make it available to any peer.

```
R1(config)#crypto isakmp key kevinskey address 0.0.0.0 0.0.0.0
```

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**Step 5:** On R1, let's create a Transform set for our IKE Phase 2 connection. In this lab we are going to be using transport mode. Name our transform set "KWTRAIN".

```
R1(config)#crypto ipsec transform-set KWTRAIN ?
  ah-md5-hmac      AH-HMAC-MD5 transform
  ah-sha-hmac      AH-HMAC-SHA transform
  ah-sha256-hmac   AH-HMAC-SHA256 transform
  ah-sha384-hmac   AH-HMAC-SHA384 transform
  ah-sha512-hmac   AH-HMAC-SHA512 transform
  comp-lzs         IP Compression using the LZS compression algorithm
  esp-3des         ESP transform using 3DES(EDE) cipher (168 bits)
  esp-aes          ESP transform using AES cipher
  esp-des          ESP transform using DES cipher (56 bits)
  esp-gcm          ESP transform using GCM cipher
  esp-gmac         ESP transform using GMAC cipher
  esp-md5-hmac     ESP transform using HMAC-MD5 auth
  esp-null         ESP transform w/o cipher
  esp-seal         ESP transform using SEAL cipher (160 bits)
  esp-sha-hmac     ESP transform using HMAC-SHA auth
  esp-sha256-hmac  ESP transform using HMAC-SHA256 auth
  esp-sha384-hmac  ESP transform using HMAC-SHA384 auth
  esp-sha512-hmac  ESP transform using HMAC-SHA512 auth

R1(config)#crypto ipsec transform-set KWTRAIN esp-aes ?
  128              128 bit keys.
  192              192 bit keys.
  256              256 bit keys.
  ah-md5-hmac      AH-HMAC-MD5 transform
  ah-sha-hmac      AH-HMAC-SHA transform
  ah-sha256-hmac   AH-HMAC-SHA256 transform
  ah-sha384-hmac   AH-HMAC-SHA384 transform
  ah-sha512-hmac   AH-HMAC-SHA512 transform
  comp-lzs         IP Compression using the LZS compression algorithm
  esp-md5-hmac     ESP transform using HMAC-MD5 auth
  esp-sha-hmac     ESP transform using HMAC-SHA auth
  esp-sha256-hmac  ESP transform using HMAC-SHA256 auth
  esp-sha384-hmac  ESP transform using HMAC-SHA384 auth
  esp-sha512-hmac  ESP transform using HMAC-SHA512 auth
  <cr>             <cr>

R1(config)#crypto ipsec transform-set KWTRAIN esp-aes esp-sha-hmac
R1(cfg-crypto-trans)#mode ?
  transport  transport (payload encapsulation) mode
  tunnel     tunnel (datagram encapsulation) mode
```

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```
R1(cfg-crypto-trans)#mode transport
R1(cfg-crypto-trans)#exit
```

*(##Here is where we are indicating which security algorithm we are going to be using.)*
*(##Here is where we are indicating how we are going to be doing hashing.)*

**Step 6:** Let's define our "interesting" traffic for the IPsec tunnel, and we want it to be our GRE traffic. Name our access control list "GRE-IN-IPSEC".

```
R1(config)#ip access-list extended GRE-IN-IPSEC
R1(config-ext-nacl)#permit gre any any
R1(config-ext-nacl)#exit
```

**Step 7:** Let's create a Crypto Map and name it "VPN". This is going to be the thing that ties everything together, the interesting traffic, the transform set, and we point all this to our peer.

```
R1(config)#crypto map VPN ?
  <1-65535>       Sequence to insert into crypto map entry
  client          Specify client configuration settings
  gdoi            Configure crypto map gdoi features
  isakmp          Specify isakmp configuration settings
  isakmp-profile  Specify isakmp profile to use
  local-address   Interface to use for local address for this crypto map
  redundancy      High availability options for this map

R1(config)#crypto map VPN 10 ?
  gdoi          configure GDOI map
  ipsec-isakmp  IPSEC w/ISAKMP
  ipsec-manual  IPSEC w/manual keying
  <cr>          <cr>

R1(config)#crypto map VPN 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
        and a valid access list have been configured.
R1(config-crypto-map)#match address GRE-IN-IPSEC
R1(config-crypto-map)#set transform-set KWTRAIN
R1(config-crypto-map)#set peer 198.51.100.2
R1(config-crypto-map)#exit
```

*(##This is for an IPsec tunnel that uses isakmp, which is what we typically use.)*
*(##Notice how this is going to be disabled until we specify a valid peer and a valid access-list.)*

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*(##Here we are specifying the transform set that we want to use to protect the traffic.  Any traffic that matches this access-list, we are going to put that into the IPsec tunnel.)*
*(##Our peer is the incoming interface on R4.)*

**Step 8:** We now need to apply this crypto map to the gig 0/1 interface, which is the interface that we are going to use to send the GRE packets.

```
R1(config)#int gig 0/1
R1(config-if)#crypto map VPN
R1(config-if)#exit
```

**Step 9:** Let's now go to R4 and give a mirrored configuration of R1.

```
R4#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R4(config)#crypto isakmp policy 10
R4(config-isakmp)#encryption aes
R4(config-isakmp)#authentication pre-share
R4(config-isakmp)#group 2
R4(config-isakmp)#exit
R4(config)#crypto isakmp key kevinskey address 0.0.0.0 0.0.0.0
R4(config)#crypto ipsec transform-set KWTRAIN esp-aes esp-sha-hmac
R4(cfg-crypto-trans)#mode transport
R4(cfg-crypto-trans)#exit
R4(config)#ip access-list extended GRE-IN-IPSEC
R4(config-ext-nacl)#permit gre any any
R4(config-ext-nacl)#exit
R4(config)#crypto map VPN 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
        and a valid access list have been configured.
R4(config-crypto-map)#match address GRE-IN-IPSEC
R4(config-crypto-map)#set transform KWTRAIN
R4(config-crypto-map)#set peer 192.0.2.1
R4(config-crypto-map)#exit
R4(config)#int gig 0/1
R4(config-if)#crypto map VPN
R4(config-if)#end
```

**Step 10:** On R4, let's do some configuration verification by sending some traffic over the tunnel. Let's see if the IPsec tunnel is encrypting and decrypting GRE packets (our only interesting traffic).

```
R4#ping 1.1.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 1.1.1.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/11/17 ms
R4#show crypto isakmp sa
```

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```
IPv4 Crypto ISAKMP SA
dst             src             state          conn-id status
198.51.100.2    192.0.2.1       QM_IDLE           1001 ACTIVE

IPv6 Crypto ISAKMP SA

R4#show crypto ipsec sa

interface: GigabitEthernet0/1
    Crypto map tag: VPN, local addr 198.51.100.2

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/47/0)
   remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/47/0)
   current_peer 192.0.2.1 port 500
     PERMIT, flags={origin_is_acl,}
    #pkts encaps: 5396, #pkts encrypt: 5396, #pkts digest: 5396
    #pkts decaps: 5395, #pkts decrypt: 5395, #pkts verify: 5395
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0

     local crypto endpt.: 198.51.100.2, remote crypto endpt.: 192.0.2.1
     plaintext mtu 1438, path mtu 1500, ip mtu 1500, ip mtu idb
GigabitEthernet0/1
     current outbound spi: 0x1B935A4B(462641739)
     PFS (Y/N): N, DH group: none

     inbound esp sas:
      spi: 0x25A4AAE9(631548649)
        transform: esp-aes esp-sha-hmac ,
        in use settings ={Tunnel, }
        conn id: 1, flow_id: SW:1, sibling_flags 80000040, crypto map: VPN
        sa timing: remaining key lifetime (k/sec): (4324133/3191)
        IV size: 16 bytes
        replay detection support: Y
        Status: ACTIVE(ACTIVE)

     inbound ah sas:

     inbound pcp sas:

     outbound esp sas:
      spi: 0x1B935A4B(462641739)
        transform: esp-aes esp-sha-hmac ,
        in use settings ={Tunnel, }
        conn id: 2, flow_id: SW:2, sibling_flags 80000040, crypto map: VPN
        sa timing: remaining key lifetime (k/sec): (4324133/3191)
        IV size: 16 bytes
        replay detection support: Y
        Status: ACTIVE(ACTIVE)

     outbound ah sas:

     outbound pcp sas:
```

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*(##This is verification for IKE Phase 1, or isakmp.)*
*(##This is verification of IPsec. Notice how we can see the encapsulation and de-encapsulation of packets happening. This is confirmation that we are successfully sending GRE traffic inside of our IPsec tunnel.)*

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